

D326 Advance Data Management

Performance Assessment

Kaitlin Dobay 011885150

05/24/2026

Part A

This report will analyze the revenue generated by different film categories in the DVD rental database. My business question is “Which film category generates the most revenue overall?” I will be including 2 tables: the detailed table with all DVD individual information, and a summary table that shows the total revenue broken down by film category. The goal of this report is to show which categories are doing well and which are not. Once shown, marketing, inventory, and focus can be shifted to either focus on well-performing categories or focus on the categories that need more attention.

Part A1

The detailed table will include customer_id, customer_name (more information in Step A4 and calculated in Step B), film_id, category_name, rental_date, and payment_amount.

The summary table will include category_name and total revenue.

Part A2

The detailed table will show you the data types INTEGER, TEXT, VARCHAR, TIMESTAMP, and NUMERIC.

The summary table will include VARCHAR and NUMERIC.

Part A3

The fields for the summary table will be pulled from the detailed table.

The detailed table will pull information from the following tables: customer, film, category, rental, and payment.

Part A4

One field on the detailed table that will be a calculation will be the customer_name. We do not need their name divided into first and last names, and we will be calculating them together.

This is for readability for those reviewing my report and findings.

Part A5

The detailed table gives the data on the level of each transaction on a minimal clean level, including the rental information, customer information, and basic film information. This gives us the ability to analyze customer trends, specific film trends, and overall specifics. This is used for a more in-depth and detailed analysis.

The summary table is an aggregated outlook of information. This shows the total revenue of films by category. This gives important insights into how the genre of movies is actually doing, which is doing well on its own with additional marketing and strategies, or what categories need more help, whether it is actor selection, more marketing, or just not worth pursuing any longer. This table is for making more strategic decisions long-term.

Part A6

This report should be run monthly for comparison. Monthly data allows those studying trends to have enough data to make future predictions with marketing, strategy, and inventory.

Part B

-- drop older versions

DROP FUNCTION IF EXIST format_customer_name (VARCHAR, VARCHAR);

DROP FUNCTION IF EXIST format_customer_name (TEXT, TEXT);

-- successful.

--create function

CREATE OR REPLACE FUNCTION

format_customer_name(p_first_name TEXT, p_last_name TEXT)

RETURNS TEXT

LANGUAGE plpgsql

AS \$func\$

BEGIN

RETURN TRIM(p_first_name) || ' ' || TRIM(p_last_name);

END;

\$func\$;

--successful

--test

SELECT format_customer_name('Mary', 'Smith') AS test_name;

--successful

Part C

--part C

--drop previous tables

DROP TABLE IF EXISTS detailed_revenue_by_category;

DROP TABLE IF EXISTS summary_revenue_by_category;

--successful

-- create detailed table

CREATE TABLE detailed_revenue_by_category (

detail_id SERIAL PRIMARY KEY,

customer_id INTEGER NOT NULL,

customer_name TEXT NOT NULL,

film_id INTEGER NOT NULL,

category_name TEXT NOT NULL,

rental_date TIMESTAMP NOT NULL,

payment_amount NUMERIC(6,2) NOT NULL

);

--successful

-- Summary table (aggregated answer to business question)

CREATE TABLE summary_revenue_by_category (

category_name TEXT PRIMARY KEY,

total_revenue NUMERIC(12,2) NOT NULL

);

Part D

```
SELECT
c.customer_id,
format_customer_name(c.first_name, c.last_name) AS customer_name,
i.film_id,
cat.name AS category_name,
r.rental_date,
p.amount AS payment_amount
FROM payment p
JOIN rental AS r
ON p.rental_id = r.rental_id
JOIN customer AS c
ON r.customer_id = c.customer_id
JOIN inventory AS i
ON r.inventory_id = i.inventory_id
JOIN film_category AS fc
ON i.film_id = fc.film_id
JOIN category AS cat
ON fc.category_id = cat.category_id;
--successful
```

Part E1

```
--create trigger function
CREATE OR REPLACE FUNCTION trg_update_summary_revenue()
RETURNS TRIGGER
LANGUAGE plpgsql
AS $$
BEGIN
INSERT INTO summary_revenue_by_category (category_name, total_revenue)
VALUES (
NEW.category_name,
(SELECT COALESCE(SUM(payment_amount), 0)
FROM detailed_revenue_by_category
WHERE category_name = NEW.category_name)
)
ON CONFLICT (category_name)
DO UPDATE
SET total_revenue = EXCLUDED.total_revenue;

RETURN NEW;
END;
$$;
--successful
```

Part E2

```
--create Trigger
DROP TRIGGER IF EXISTS trg_after_insert_update_summary
```

```
ON detailed_revenue_by_category;
```

```
CREATE TRIGGER trg_after_insert_update_summary  
AFTER INSERT ON detailed_revenue_by_category  
FOR EACH ROW  
EXECUTE FUNCTION trg_update_summary_revenue();  
--successful
```

```
--test trigger  
SELECT * FROM summary_revenue_by_category;  
--successful
```

Part F

```
--create procedure  
CREATE OR REPLACE PROCEDURE refresh_revenue_report()  
LANGUAGE plpgsql  
AS $$  
BEGIN  
TRUNCATE TABLE summary_revenue_by_category;  
TRUNCATE TABLE detailed_revenue_by_category;  
INSERT INTO detailed_revenue_by_category  
(customer_id, customer_name, film_id, category_name, rental_date, payment_amount)  
SELECT  
c.customer_id,  
format_customer_name(c.first_name, c.last_name) AS customer_name,  
i.film_id,  
cat.name AS category_name,  
r.rental_date,  
p.amount AS payment_amount  
FROM payment p  
JOIN rental AS r  
ON p.rental_id = r.rental_id  
JOIN customer AS c  
ON r.customer_id = c.customer_id
```

```
JOIN inventory AS i
ON r.inventory_id = i.inventory_id
JOIN film_category AS fc
ON i.film_id = fc.film_id
JOIN category cat
ON fc.category_id = cat.category_id;
END;
$$;
--successful
```

Part F1

I do recommend using PgAdmin for scheduling. It has the capabilities to store procedures, schedule reports to be run, and is designed for PostgreSQL. Manual execution is not necessary for PgAdmin to run this planned code.

Part G

Here is the link for the report overview and Live code.

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=d02558ba-f18e-429b-9f71-b45401025832>

Part H

No sources were used in the creation of this assessment.